

Bass Diffusion Model: Jackery Solar Roof

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Importing libraries

```
library(ggplot2)
library(readxl)
library(diffusion)
```

Loading the dataset

```
data <- read_excel("data/cumulative_solar_pv.xlsx", sheet = "Data", col_names = TRUE)
head(data)
```

```
## # A tibble: 6 x 2
##   Year Capacity_MW
##   <dbl>         <dbl>
## 1  2000           1288
## 2  2001           1615
## 3  2002           2069
## 4  2003           2635
## 5  2004           3723
## 6  2005           5112
```

Data preparation

```
capacity <- data$Capacity_MW

# Converting cumulative capacity to yearly adoption
sales <- diff(capacity)

# Time index
t <- 1:length(sales)

sales
```

```
## [1] 327 454 566 1088 1389 1548 2523 6661 7341 17151
## [11] 30133 30035 38352 39535 50909 77200 98000 104800 124400 138500
## [21] 167800 237000 447000 576000
```

STEP 4 - Estimating Bass Model Parameters

Bass model function

```
bass_sales <- function(t, p, q, m) {
  m * ((p + q)^2 / p) * exp(-(p + q) * t) / (1 + (q / p) * exp(-(p + q) * t))^2
}
```

Method 1 - Estimating Bass Parameters using nls()

```
bass_model <- nls(
  sales ~ bass_sales(t, p, q, m),
  start = list(m = max(capacity) * 1.5, p = 0.001, q = 0.2),
  control = nls.control(maxiter = 1000, minFactor = 1/2048, warnOnly = TRUE)
)
```

```
## Warning in nls(sales ~ bass_sales(t, p, q, m), start = list(m = max(capacity) *
## : step factor 0.000244141 reduced below 'minFactor' of 0.000488281
```

```
summary(bass_model)
```

```
##
## Formula: sales ~ bass_sales(t, p, q, m)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## m 3.124e+07  1.587e+09   0.020   0.984
## p 1.225e-04  5.758e-03   0.021   0.983
## q 1.639e-01  3.164e-01   0.518   0.610
##
## Residual standard error: 111300 on 21 degrees of freedom
##
## Number of iterations till stop: 41
## Achieved convergence tolerance: 4.946
## Reason stopped: step factor 0.000244141 reduced below 'minFactor' of 0.000488281
```

```
# Extracting parameters
p_est <- coef(bass_model)["p"]
q_est <- coef(bass_model)["q"]
m_est <- coef(bass_model)["m"]

p_est
```

```
##
##           p
## 0.0001224869
```

```
q_est
```

```
##
##           q
## 0.1638765
```

```
m_est
```

```
##
##           m
## 31240030
```

Method 2 - Estimating Bass Parameters using diffusion library

```
diff_m <- diffusion(sales, type = "bass")
```

```
## Warning in diffusion(sales, type = "bass"): Initialisation parameters are of  
## different scale. Optimization might be impacted
```

```
# Extracting and rounding parameters
```

```
p_diff <- round(diff_m$w["p"], 4)
```

```
q_diff <- round(diff_m$w["q"], 4)
```

```
m_diff <- round(diff_m$w["m"], 0)
```

```
diff_m
```

```
## bass model
```

```
##
```

```
## Parameters:
```

```
##      Estimate p-value
```

```
## m 7149665.2930      NA
```

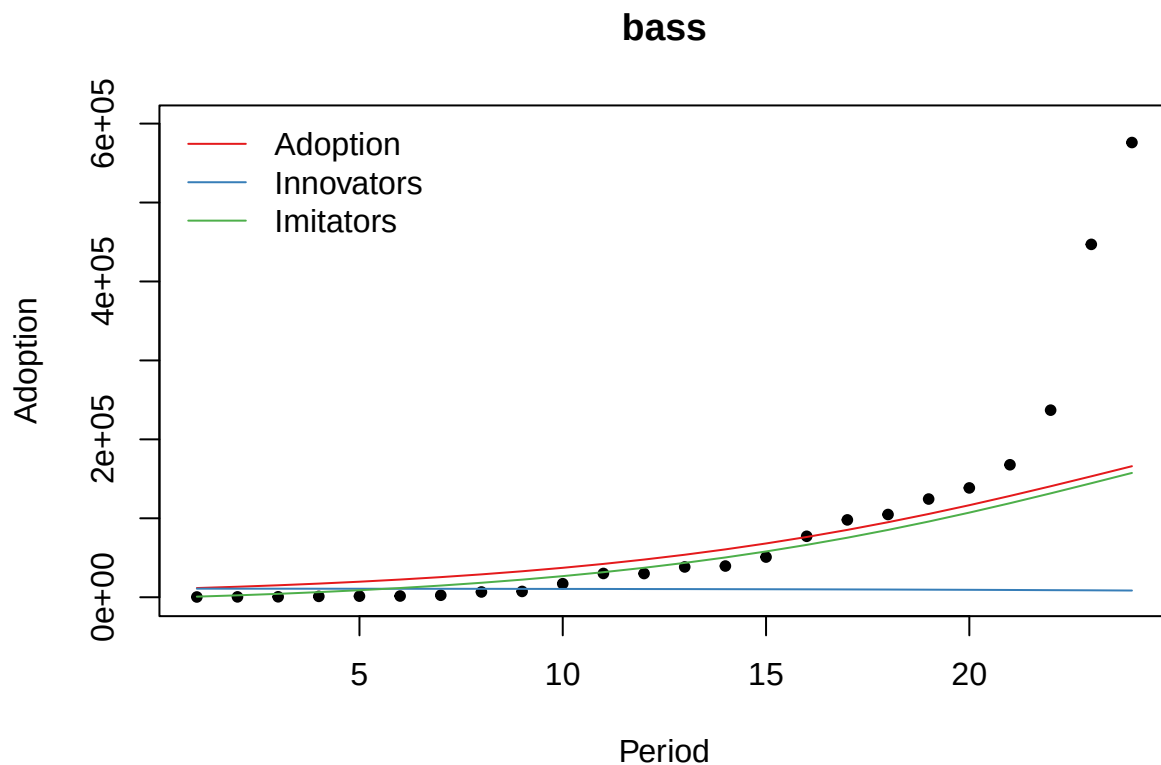
```
## p      0.0015      NA
```

```
## q      0.1341      NA
```

```
##
```

```
## sigma: 106350.8779
```

```
plot(diff_m)
```



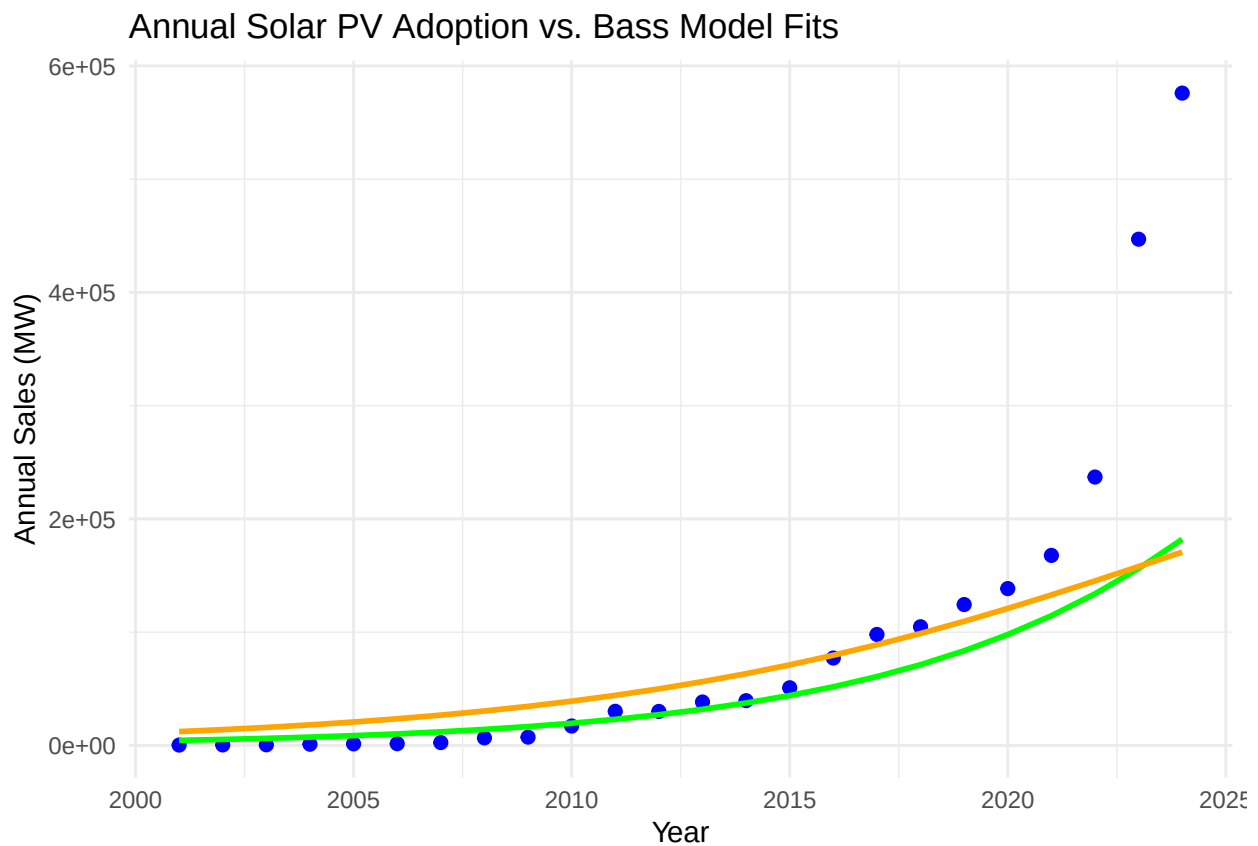
Dataframe for comparing the methods

```
# fitted annual sales
fitted_nls <- bass_sales(t, p_est, q_est, m_est)
fitted_diff <- bass_sales(t, p_diff, q_diff, m_diff)

df1 <- data.frame(
  Year = 2000 + t,
  Actual = sales,
  NLS_Fit = fitted_nls,
  DIFF_Fit = fitted_diff
)
```

Annual Sales vs Diffusion

```
ggplot(df1, aes(x = Year)) +
  geom_point(aes(y = Actual), color = "blue", size = 2) +
  geom_line(aes(y = NLS_Fit), color = "green", linewidth = 1) +
  geom_line(aes(y = DIFF_Fit), color = "orange", linewidth = 1) +
  labs(title = "Annual Solar PV Adoption vs. Bass Model Fits",
        y = "Annual Sales (MW)") +
  theme_minimal()
```



STEP 5 - Predicting Diffusion Path

```
bass_cum <- function(t, p, q, m) {
  m * (1 - exp(-(p + q) * t)) / (1 + (q / p) * exp(-(p + q) * t))
}
```

Forecast for the next 30 years using

```
t_forecast <- 1:35

# Generating predictions
sales_pred_nls <- bass_sales(t_forecast, p_est, q_est, m_est)
sales_pred_diff <- bass_sales(t_forecast, p_diff, q_diff, m_diff)

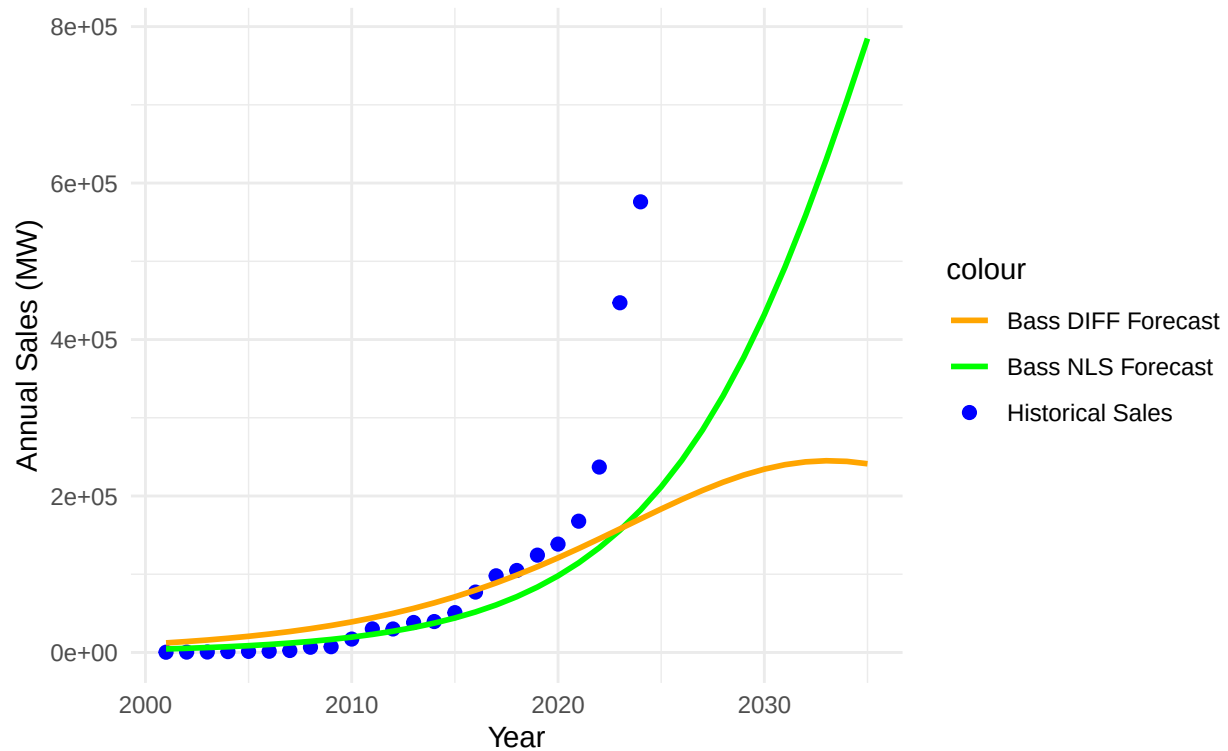
# Prediction dataframe
df_forecast <- data.frame(
  Year = 2000 + t_forecast,
  NLS_Predicted_Sales = sales_pred_nls,
  DIFF_Predicted_Sales = sales_pred_diff
)
```

Plotting historical vs predicted diffusion

```
ggplot() +
  geom_point(data = data.frame(Year = 2000 + t, Actual = sales),
    aes(x = Year, y = Actual, color = "Historical Sales"), size = 2) +
  geom_line(data = df_forecast,
    aes(x = Year, y = NLS_Predicted_Sales, color = "Bass NLS Forecast"), linewidth = 1) +
  geom_line(data = df_forecast,
    aes(x = Year, y = DIFF_Predicted_Sales, color = "Bass DIFF Forecast"), linewidth = 1) +
  scale_color_manual(values = c("Historical Sales" = "blue", "Bass NLS Forecast" = "green",
    "Bass DIFF Forecast" = "orange")) +
  labs(title = "Annual Adoption Forecast: Jackery Solar Roof",
    subtitle = "Projection of the Imitation effect through 2035",
    y = "Annual Sales (MW)", x = "Year") +
  theme_minimal()
```

Annual Adoption Forecast: Jackery Solar Roof

Projection of the Imitation effect through 2035



STEP 6 AND 7 - Using Netherlands Residential Solar PV Data

Load the dataset

```
nl_data <- read_excel("data/netherlands_pv.xlsx", sheet = "Data", col_names = TRUE)
head(nl_data)
```

```
## # A tibble: 6 x 2
##   Year PV_Systems
##   <dbl>     <dbl>
## 1  2019     960248
## 2  2020    1267651
## 3  2021    1611689
## 4  2022    2131019
## 5  2023    2654864
## 6  2024    2790371
```

Prepare data for Bass model

```
annual_sales <- diff(nl_data$PV_Systems)
t <- 1:length(annual_sales)
```

Diffusion Method

```
diff_m <- diffusion(annual_sales, type = "bass")
```

```
# Extracting and rounding parameters
```

```
p_diff <- round(diff_m$w["p"], 4)
```

```
q_diff <- round(diff_m$w["q"], 4)
```

```
m_diff <- round(diff_m$w["m"], 0)
```

```
diff_m
```

```
## bass model
```

```
##
```

```
## Parameters:
```

```
##      Estimate p-value
```

```
## m 2045078.9536      NA
```

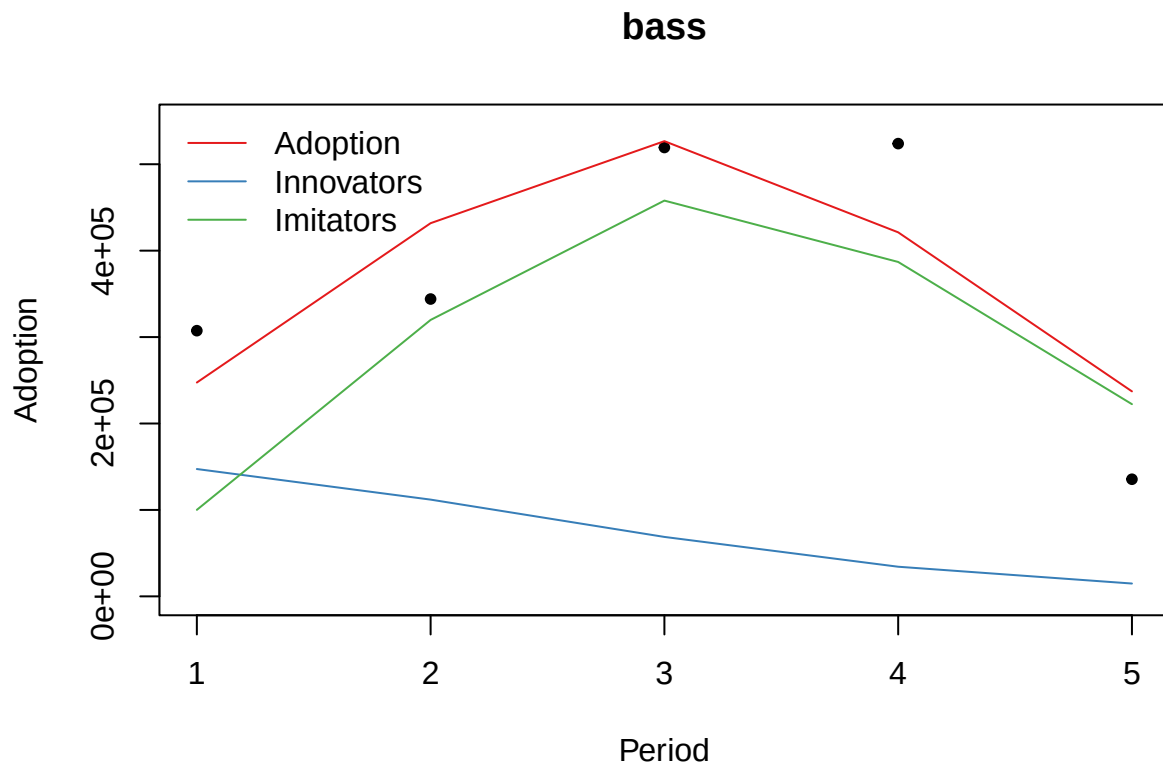
```
## p      0.0820      NA
```

```
## q      0.8785      NA
```

```
##
```

```
## sigma: 80282.1157
```

```
plot(diff_m)
```



Estimate Bass parameters using nls()

```
bass_model <- nls(
  annual_sales ~ bass_sales(t, p, q, m),
```

```

start = list(m = max(nl_data$PV_Systems) * 1.5, p = 0.01, q = 0.3),
control = nls.control(maxiter = 1000, minFactor = 1/2048, warnOnly = TRUE)
)

summary(bass_model)

```

```

##
## Formula: annual_sales ~ bass_sales(t, p, q, m)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## m 2.049e+06  4.011e+05   5.109  0.0362 *
## p 4.588e-02  2.983e-02   1.538  0.2639
## q 9.512e-01  3.257e-01   2.921  0.1000 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 125100 on 2 degrees of freedom
##
## Number of iterations to convergence: 31
## Achieved convergence tolerance: 4.17e-06

```

```

# Extracting estimated parameters
p_est <- coef(bass_model)["p"]
q_est <- coef(bass_model)["q"]
m_est <- coef(bass_model)["m"]

p_est

```

```

##           p
## 0.04587575

```

```
q_est
```

```

##           q
## 0.951175

```

```
m_est
```

```

##           m
## 2048882

```

Forecasting future adoption (2019-2030)

```

t_future <- 1:12
cum_pred <- bass_cum(t_future, p_est, q_est, m_est)
annual_pred <- c(cum_pred[1], diff(cum_pred))

```

Dataframe for plotting

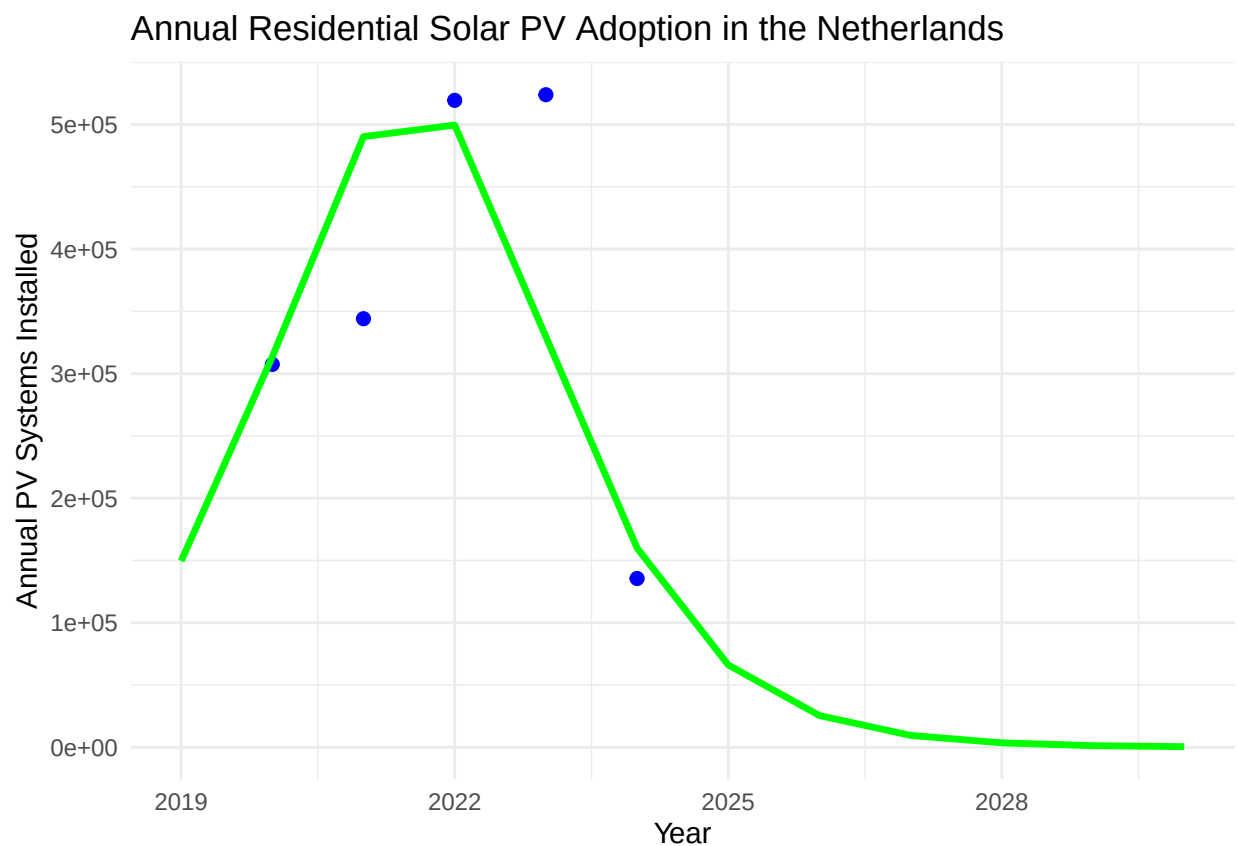

```
df_pred <- data.frame(
  Year = 2019 + t_future - 1,
  Predicted_Annual = annual_pred,
  Predicted_Cumulative = cum_pred
)
```

Add historical data for comparison

```
df_hist <- data.frame(
  Year = nl_data$Year[-1], # removing first year since diff() reduces length
  Actual_Annual = annual_sales,
  Actual_Cumulative = nl_data$PV_Systems[-1]
)
```

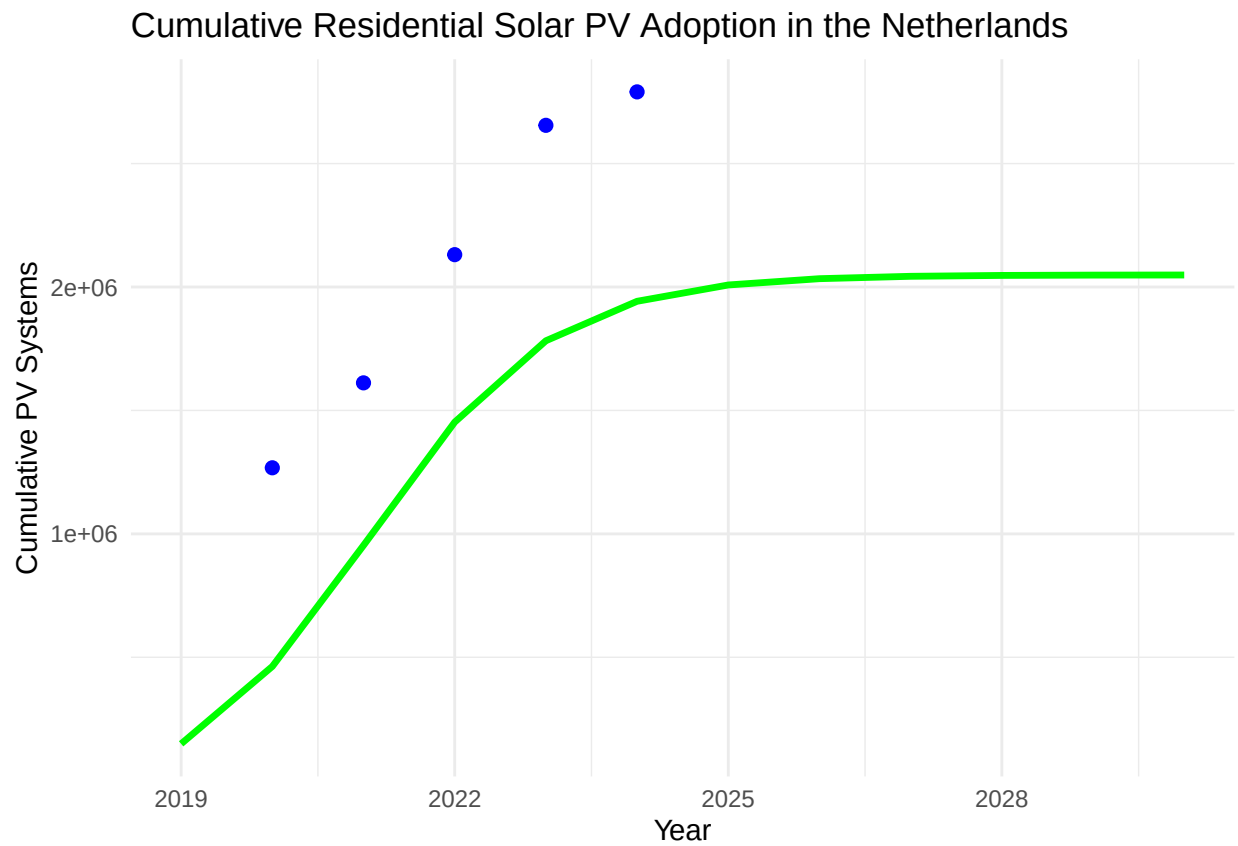
Plotting annual adoption: historical vs predicted

```
ggplot() +
  geom_point(data = df_hist, aes(x = Year, y = Actual_Annual), color = "blue", size = 2) +
  geom_line(data = df_pred, aes(x = Year, y = Predicted_Annual), color = "green", linewidth = 1.2) +
  labs(title = "Annual Residential Solar PV Adoption in the Netherlands",
       y = "Annual PV Systems Installed",
       x = "Year") +
  theme_minimal()
```



Plotting cumulative adoption: historical vs predicted

```
ggplot() +
  geom_point(data = df_hist, aes(x = Year, y = Actual_Cumulative), color = "blue", size = 2) +
  geom_line(data = df_pred, aes(x = Year, y = Predicted_Cumulative), color = "green", linewidth = 1.2) +
  labs(title = "Cumulative Residential Solar PV Adoption in the Netherlands",
       y = "Cumulative PV Systems",
       x = "Year") +
  theme_minimal()
```



Computing the time of peak adoption

```
t_peak <- log(q_est / p_est) / (p_est + q_est)
peak_year <- 2019 + t_peak - 1
peak_year
```

```
##          q
## 2021.041
```

FERMI LOGIC

Estimating number of adopters by period using Fermi calculation ($m = 3.27$ million)

```
m_fermi <- 3272000

cum_pred_fermi <- bass_cum(t_future, p_est, q_est, m_fermi)
annual_pred_fermi <- c(cum_pred_fermi[1], diff(cum_pred_fermi))
```

```
df_pred_fermi <- data.frame(
  Year = 2019 + t_future - 1,
  Predicted_Annual = annual_pred_fermi,
  Predicted_Cumulative = cum_pred_fermi
)
```

Plotting cumulative adoption: historical vs predicted

```
ggplot() +
  geom_point(data = df_hist, aes(x = Year, y = Actual_Cumulative), color = "blue", size = 2) +
  geom_line(data = df_pred_fermi, aes(x = Year, y = Predicted_Cumulative), color = "black", linewidth = 2) +
  labs(title = "Fermi Cumulative Residential Solar PV Adoption in the Netherlands",
       y = "Cumulative PV Systems",
       x = "Year") +
  theme_minimal()
```

